

Trigonometric functions and equations - Set 2

TMUA practice questions by topic. Specification MM4.3-MM4.6. 7 questions.

Each question has one correct answer. No calculator. Answers and worked solutions are in the separate solutions PDF.

- 1 Consider the equation $\sin x = k$ where k is a real constant. Which of the following statements is/are true?

- I For every k with $-1 < k < 1$ the equation has exactly two solutions in $0 \leq x < 2\pi$.
- II For $k = 1$ the equation has exactly one solution in $0 \leq x < 2\pi$.
- III For every real k the equation has at least one solution.

- A none of them
- B I only
- C II only
- D III only
- E I and II only
- F I and III only
- G II and III only
- H I, II and III

- 2 Solve $2\cos^2 x + 3\sin x - 3 = 0$ for $0^\circ \leq x \leq 360^\circ$. How many solutions are there?

- A 2
- B 0
- C 4
- D 5
- E 3
- F 1

- 3 Which of the following statements about the graph of $y = \cos x$ is/are true?
- I It is symmetric about the y -axis.
 - II It is a translation of the graph of $y = \sin x$ by $\frac{\pi}{2}$ in the negative x -direction.
 - III It has period π .
- A none of them
- B I only
- C II only
- D III only
- E I and II only
- F I and III only
- G II and III only
- H I, II and III
- 4 Solve $12\cos^2 x + 6\sin x - 10 = 2$ for $0^\circ < x < 360^\circ$. Which of the following is the complete set of solutions?
- A $0^\circ, 30^\circ, 150^\circ, 180^\circ$
- B $30^\circ, 150^\circ, 210^\circ, 330^\circ$
- C $30^\circ, 150^\circ, 270^\circ$
- D $30^\circ, 150^\circ$
- E $210^\circ, 330^\circ$
- F $30^\circ, 150^\circ, 180^\circ$
- 5 The graph of $y = \sin x$ (with x in radians) crosses the line $y = x$ at how many points?
- A 2
- B 3
- C 0
- D 1
- E infinitely many

6 Which of the following is/are true for all real x ?

I $\sin^2 x + \cos^2 x = 1$

II $\sin(2x) = 2\sin x$

III $\sin(x + \pi) = -\sin x$

A none of them

B I only

C II only

D III only

E I and II only

F I and III only

G II and III only

H I, II and III

7 How many solutions does the equation $\sin x = \frac{x}{10}$ have?

A 7

B 6

C 1

D 5

E 9

F 3